SI Units and Unit Conversions

Answer key with worked solutions

A. Units and prefixes

1. SI base units.

> (a) kilogram, **kg** (b) metre, **m** (c) second, **s** (d) kelvin, **K** (e) mole, **mol**

2. Powers of ten.

> (a) 1 μg = 10^{-6} g (b) 1 GHz = 10^{9} Hz (c) 1 nm = 10^{-9} m

B. One-step conversions

3. Convert.

> (a) $2.5\ \cancel{\text{km}} \times \dfrac{1000\ \text{m}}{1\ \cancel{\text{km}}} = \mathbf{2.5 \times 10^{3}\ m}$ (2500 m)
>
> (b) $350\ \cancel{\text{mL}} \times \dfrac{1\ \text{L}}{1000\ \cancel{\text{mL}}} = \mathbf{0.350\ L}$
>
> (c) $45\ \cancel{\text{mg}} \times \dfrac{1\ \text{g}}{1000\ \cancel{\text{mg}}} = \mathbf{0.045\ g}$
>
> (d) $0.0045\ \cancel{\text{kg}} \times \dfrac{1000\ \cancel{\text{g}}}{1\ \cancel{\text{kg}}} \times \dfrac{1000\ \text{mg}}{1\ \cancel{\text{g}}} = \mathbf{4.5 \times 10^{3}\ mg}$
>
> (e) $500.\ \cancel{\text{nm}} \times \dfrac{10^{-9}\ \text{m}}{1\ \cancel{\text{nm}}} = \mathbf{5.00 \times 10^{-7}\ m}$

C. Areas and volumes

4. Convert.

> (a) $5.0\ \cancel{\text{m}^2} \times \dfrac{10^{4}\ \text{cm}^2}{1\ \cancel{\text{m}^2}} = \mathbf{5.0 \times 10^{4}\ cm^2}$ (because 1 m^2 = (100 cm)^2)
>
> (b) $250\ \cancel{\text{cm}^3} \times \dfrac{1\ \text{m}^3}{10^{6}\ \cancel{\text{cm}^3}} = \mathbf{2.50 \times 10^{-4}\ m^3}$
>
> (c) $1.0\ \cancel{\text{m}^3} \times \dfrac{1000\ \text{L}}{1\ \cancel{\text{m}^3}} = \mathbf{1.0 \times 10^{3}\ L}$ (1 m^3 = 1000 dm^3 = 1000 L)
>
> (d) $0.250\ \cancel{\text{L}} \times \dfrac{1000\ \text{cm}^3}{1\ \cancel{\text{L}}} = \mathbf{250.\ cm^3}$

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D. Multi-step conversions

5. 1.5 h.

$$1.5\,\cancel{\text{h}} \times \frac{3600\ \text{s}}{1\,\cancel{\text{h}}} = \mathbf{5.4 \times 10^3\ s}$$

6. 36 km/h.

$$\frac{36\,\cancel{\text{km}}}{1\,\cancel{\text{h}}} \times \frac{1000\ \text{m}}{1\,\cancel{\text{km}}} \times \frac{1\,\cancel{\text{h}}}{3600\ \text{s}} = \mathbf{10\ m/s}$$

7. Iron.

$$\frac{7.86\,\cancel{\text{g}}}{1\,\cancel{\text{cm}^3}} \times \frac{1\ \text{kg}}{1000\,\cancel{\text{g}}} \times \frac{10^6\,\cancel{\text{cm}^3}}{1\ \text{m}^3} = \mathbf{7.86 \times 10^3\ kg/m^3}$$

8. Mercury.

$$15.0\,\cancel{\text{mL}} \times \frac{13.6\ \text{g}}{1\,\cancel{\text{mL}}} = \mathbf{204\ g}$$

E. Challenge

9. Swimming pool.

(a) $V = 25.0\ \text{m} \times 10.0\ \text{m} \times 2.00\ \text{m} = \mathbf{500.\ m^3}$

$$500.\,\cancel{\text{m}^3} \times \frac{1000\ \text{L}}{1\,\cancel{\text{m}^3}} = \mathbf{5.00 \times 10^5\ L}$$

(b)

$$m = 5.00 \times 10^5\,\cancel{\text{L}} \times \frac{1000\,\cancel{\text{mL}}}{1\,\cancel{\text{L}}} \times \frac{1.00\,\cancel{\text{g}}}{1\,\cancel{\text{mL}}} \times \frac{1\ \text{kg}}{1000\,\cancel{\text{g}}} = \mathbf{5.00 \times 10^5\ kg}$$

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